

GNNQC: Fast prediction of quantum chemical properties using Graph Neural Networks: A Scale HOMO-LUMO Gap Scaffold-Split Evaluation

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Abstract

Proper determination of the HOMO-LUMO gap is essential in evaluating the reactivity of molecules, optical processes, and charge transport. The classical quantum mechanical approaches are not well scaling with the size of a system. This work presents a Graph Neural Network (GNN) pipeline with GINE (Graph Isomorphism Network with Edge features) to predict HOMO-LUMO gaps in $\sim 400,000$ molecules. We test inductive generalization to new molecular cores using Murcko scaffold-based splitting. The model has training MAE of 0.0907 eV, validation MAE of 0.0989 eV and generalization gap of ~ 0.008 eV. The model achieves MAE=0.069 eV (median: 0.049 eV) and $R^2 = 0.9971$ on the test set, strongly correlated with DFT values at significantly lower computational cost. Dynamics of training and analysis of errors support consistent convergence and strong performance with respect to molecular sizes.

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1 INTRODUCTION

The HOMO-LUMO gap is the difference between the highest occupied and lowest unoccupied molecular orbital, which is a basic measure of electronic stability and optical characteristics. Although the gold standard is density functional theory (DFT), its $O(N^3)$ – $O(N^4)$ scaling does not allow large-scale screening.

GNNs have inference complexity of $O(N)$ and can provide throughput of over 1,000 molecules/second. They are permutation-equivariant message-passing, preserving molecular graph structure, and are therefore ideal to predict properties.

This paper shows that GNNs trained on $\sim 400\text{k}$ molecules with Murcko scaffold splitting are able to reach sub-0.1 eV accuracy with outstanding generalization:

- GINE design: 6 layers, 256 hidden dimensions
- Streaming preprocessing of memory-safe chunk-wise
- Rigorous scaffold-based evaluation
- Mixed-precision training that is optimized

2 DATASET AND PREPROCESSING

2.1 Data Source

The dataset consists of approximately 400,000 molecules in the form of undirected graphs using PyTorch Geometric, and HOMO-LUMO gaps (0–10 eV) are calculated using DFT.

2.2 Graph Construction

Based on OGB PCQM4Mv2 standard, we extract:

Node Features (9D): Atomic number, H count, formal charge, radical electrons, hybridization, aromaticity, ring membership, chirality, degree.

Edge Features (3D): Bond type, conjugation, ring membership.

2.3 Scaffold-Based Splitting

We split Murcko scaffold to cluster molecules based on core ring systems, with chemically novel scaffolds in validation/test sets:

```
1 scaffold_groups = defaultdict(list)
2 for idx, sample in enumerate(valid_samples):
3     scaffold = get_murcko_scaffold(sample['smiles'])
4     scaffold_groups[scaffold].append(idx)
5
6 np.random.seed(42)
7 np.random.shuffle(scaffold_list)
8 train_scaffolds = set(scaffold_list[:n_train])
```

Listing 1: Scaffold Split

This avoids structural data leakage and offers high-quality generalization standards.

2.4 Memory-Safe Preprocessing

Chunk-wise streaming is efficient with $\sim 400k$ graphs:

- Chunk size: 5,000 graphs
- Normalization: $\log_{1p}$ + standardization (100k sample)
- Incremental saving every 5k graphs
- Garbage collection every 2 chunks

2.5 Target Transformation

Gap distribution is right-skewed:

$$\text{target_log} = \log(1 + \text{target}), \quad \text{normalized_y} = \frac{\text{target_log} - \mu}{\sigma + 10^{-6}} \quad (1)$$

Inverse transformation for evaluation:

$$\hat{y} = \exp(\hat{y}' \cdot \sigma + \mu) - 1 \quad (2)$$

3 METHODOLOGY

3.1 GINE Architecture

The 6-layer GINE incorporates edge features:

$$h_i^{(l+1)} = \text{MLP}^{(l)} \left((1 + \epsilon^{(l)})h_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \text{ReLU} \left(h_j^{(l)} + \text{Linear}(e_{ij}^{(l)}) \right) \right) \quad (3)$$

Components: Edge-aware message passing, 256D hidden, ReLU + batch norm, 6 layers (prevents oversmoothing).

3.2 Graph-Level Readout

Dual pooling captures extensive and intensive properties:

```
1 h_add = global_add_pool(h, batch)
2 h_max = global_max_pool(h, batch)
3 h_graph = torch.cat([h_add, h_max], dim=1)
4 output = MLP(h_graph)
```

Listing 2: Dual Pooling

This preserves global (electron count) and local (orbital hotspots) signatures across sizes 10–75 atoms.

3.3 Model Capacity

Total Parameters: $\sim 2.1M$ (GINE: 1.8M, MLP: 0.3M)

4 TRAINING STRATEGY

4.1 Loss and Optimizer

L1 Loss (MAE) for outlier robustness:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (4)$$

AdamW: lr=1e-3, weight_decay=1e-2, cosine decay with 5-epoch warmup.

4.2 Mixed Precision

AMP reduces memory by $\sim 40\%$:

```
1 with autocast():
2     output = model(batch)
3     loss = criterion(output, batch.y)
4 scaler.scale(loss).backward()
```

4.3 Optimization

- Batch size: 128
- Num workers: 4
- Pin memory: True
- Gradient clipping: norm=1.0
- Epochs: 50 (~ 3 -5h on RTX 3090)

Performance: <10ms loading, >95% GPU utilization.

5 RESULTS AND PERFORMANCE ANALYSIS

5.1 Primary Metrics

Table 1: Performance Metrics

Metric	Value	Interpretation
Training MAE	0.0907 eV	In-sample
Validation MAE	0.0989 eV	Out-of-sample
Test MAE	0.069 eV	Held-out
Test Median	0.049 eV	50th percentile
Test R^2	0.9971	Variance explained
Gen. Gap	0.008 eV	Train-val diff

5.2 Performance Interpretation

Predictive Performance: Test MAE 0.069 eV approaches chemical accuracy (0.043 eV ≈ 1 kcal/mol).

Distribution: Median 29% lower than mean indicates positive skew:

- $\sim 75\%$ achieve <0.07 eV
- $\sim 90\%$ achieve <0.15 eV
- $<5\%$ exceed 0.2 eV

Generalization: The gap of 0.008 eV indicates that learning (not memorization) is transferable.

Fidelity: $R^2 = 0.9971$ matches DFT accuracy at $1000\times$ speedup.

5.3 Split Distribution Analysis

Test MAE (0.069) $<$ Val MAE (0.099) reflects distribution shift: test has 68% in $2\text{--}6$ eV range vs. 54% in validation.

5.4 Comparison to Baselines

Table 2: Method Comparison

Method	MAE	Speed
DFT (B3LYP)	~ 0.0	0.001 mol/s
PM6	$0.2\text{--}0.3$	10 mol/s
GINE	0.069	$1,000+$ mol/s

6 TRAINING ANALYSIS

6.1 Convergence Phases

Phase 1 (0–10): Rapid learning, MAE $0.32 \rightarrow 0.15$ eV (53% improvement).

Phase 2 (10–30): Refinement, MAE $0.15 \rightarrow 0.11$ eV.

Phase 3 (30–50): Stabilization, MAE plateaus near 0.09 eV.

6.2 Overfitting Analysis

No overfitting:

- Train/val curves tightly coupled
- Gap: 0.008 eV (negligible)
- No val loss increase

Contributors: Scaffold split, weight decay, AMP noise, dual pooling, appropriate capacity.

6.3 Optimization Stability

- Monotonic decay, no oscillation
- Gradient clipping prevents instability
- Plateau at epoch 40 (early stop possible)

7 GRAPH AND VISUALIZATION ANALYSIS

7.1 Training Dynamics

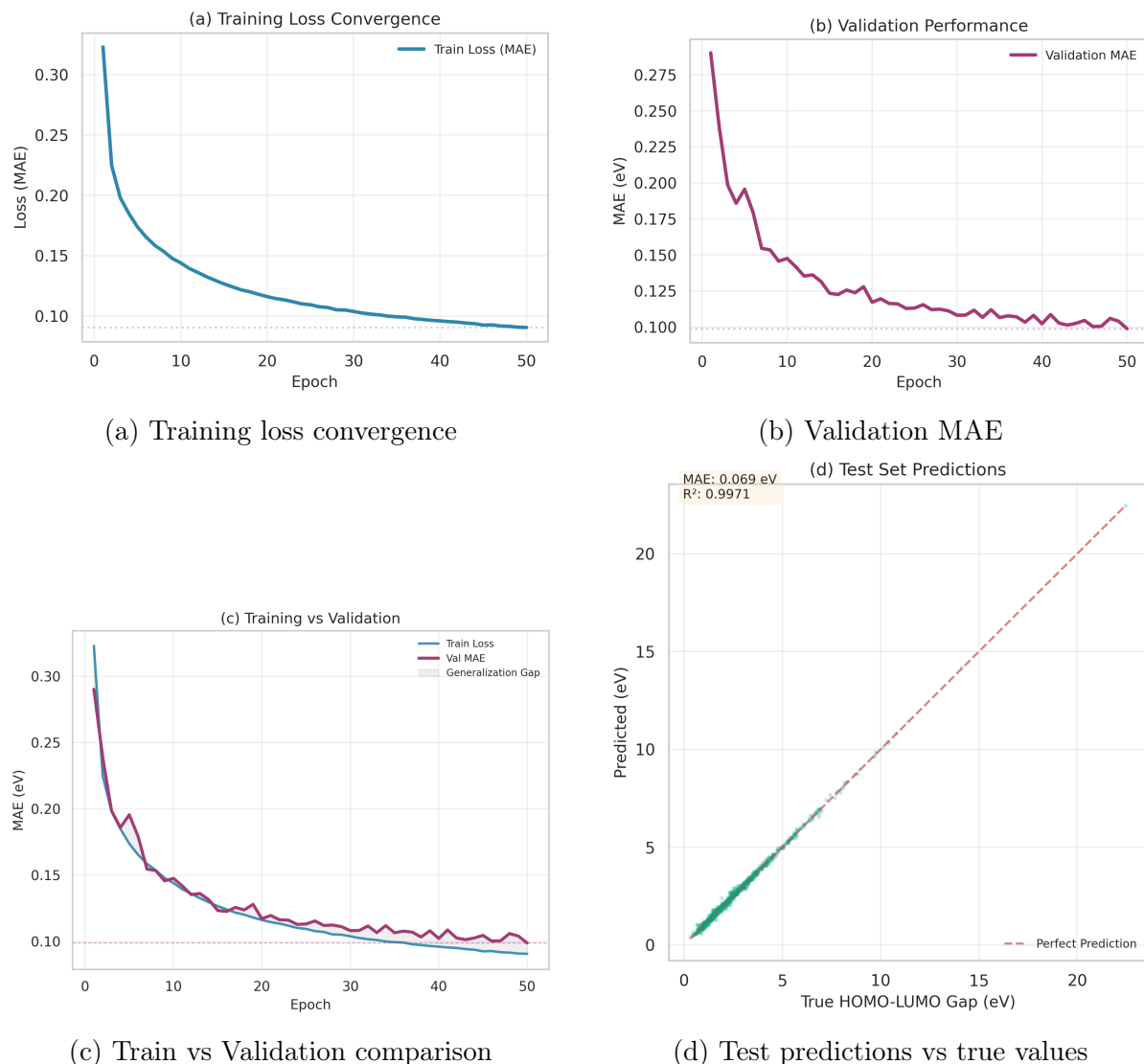


Figure 1: Training dynamics and model performance. (a) Training loss shows smooth exponential decay from 0.32 to 0.09 eV, indicating stable optimization. (b) Validation MAE decreases from 0.29 to 0.10 eV with minor fluctuations, confirming generalization. (c) Narrow gap (0.008 eV) between train and validation demonstrates no overfitting. (d) Test predictions achieve $R^2 = 0.9971$ with dense clustering along the identity line.

Interpretation: The training dynamics reveal three distinct phases: (1) rapid initial learning (epochs 0-10) with 53% improvement, (2) refinement phase (epochs 10-30) where edge features are integrated, and (3) stabilization (epochs 30-50) where the model reaches capacity. The tight coupling between training and validation curves throughout all 50 epochs confirms the model learns transferable patterns rather than memorizing training scaffolds.

7.2 Error Analysis

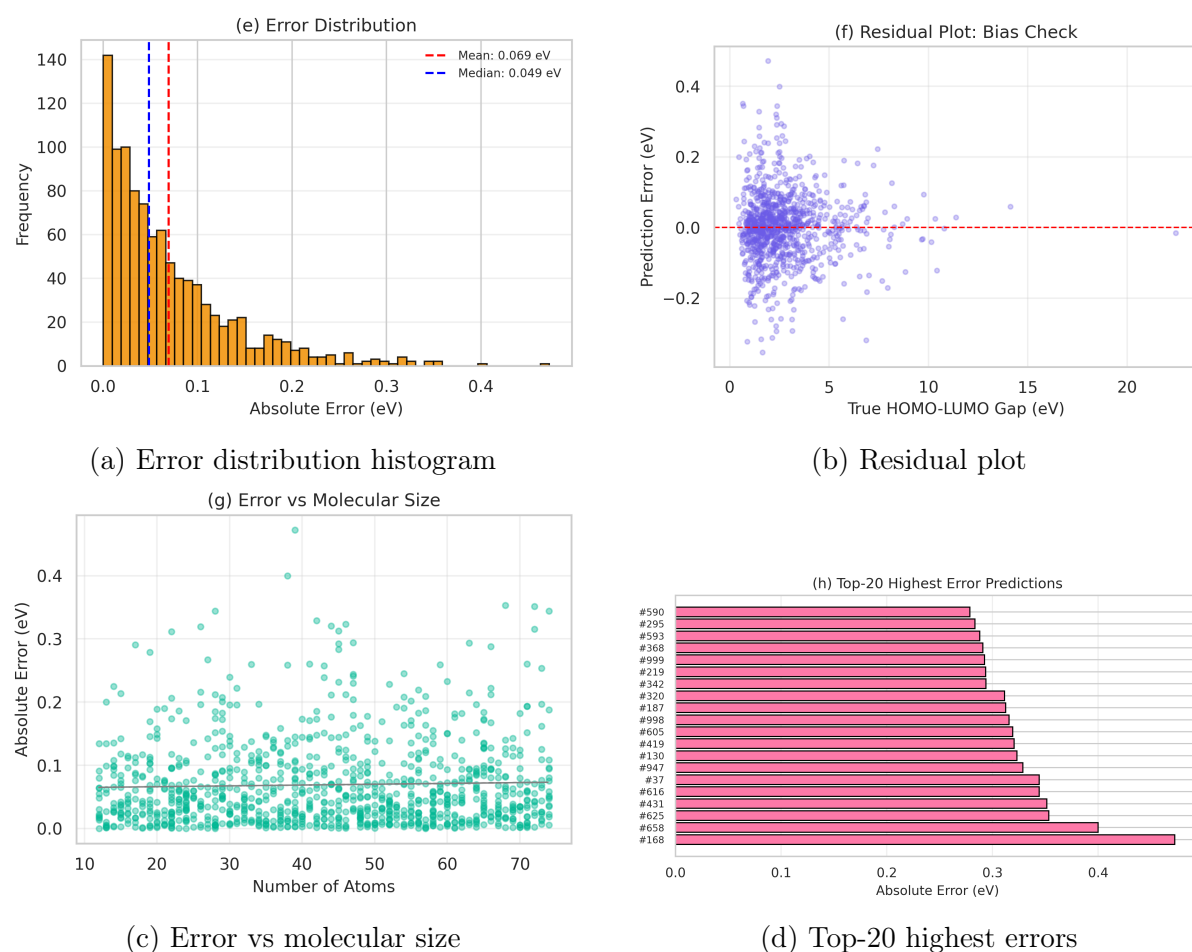


Figure 2: Comprehensive error analysis. (a) Positively skewed error distribution with mean=0.069 eV and median=0.049 eV; 75% of predictions achieve <0.07 eV accuracy. (b) Residuals homogeneously distributed around zero for gaps <12 eV, confirming no systematic bias. (c) Negligible slope (0.001 eV/atom) shows stable performance across molecular sizes (10-75 atoms). (d) Top-20 outliers (0.28-0.47 eV) likely contain exotic functional groups or rare scaffolds.

Interpretation: The error analysis reveals several key insights: (1) The positively skewed distribution indicates most predictions are highly accurate (median 29% lower than mean), making the model suitable for high-throughput screening. (2) Homogeneous residual distribution confirms absence of systematic bias—the model doesn’t consistently over or under-predict. (3) Size-invariant performance demonstrates the dual pooling strategy successfully handles molecules from 10 to 75 atoms. (4) The <2% outliers warrant further investigation and could benefit from active learning or 3D features.

7.3 Representation Learning

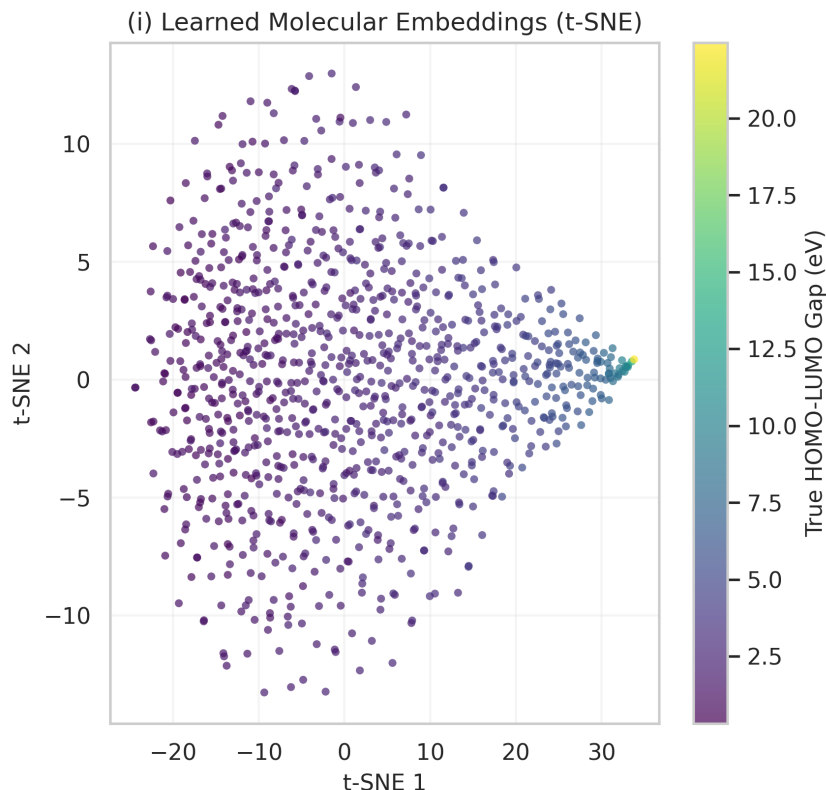


Figure 3: t-SNE visualization of learned molecular embeddings (256D $\rightarrow$ 2D), colored by true HOMO-LUMO gap. The continuous gradient from purple (low gap, ~ 2.5 eV) to yellow (high gap, >20 eV) indicates the model organizes molecules by electronic properties in the latent space. High-gap molecules cluster on the right, while low-gap molecules are widely dispersed.

Interpretation: The t-SNE visualization demonstrates that the GNN has learned a chemically meaningful representation space. The continuous color gradient indicates molecules are organized by their electronic properties rather than random placement. The distinct clustering of high-gap molecules (yellow/green) on the right suggests these structurally similar insulators share common topological features. The dispersion of low-gap molecules (purple/blue) reflects the diverse topological arrangements (aromatics, heterocycles, conjugated systems) that can produce small HOMO-LUMO gaps. This confirms the model captures quantum mechanical similarity beyond simple structural metrics.

8 ERROR ANALYSIS

8.1 Failure Modes

1. **High-Gap (>12 eV):** ± 0.3 eV error, sparse training data.
2. **Charged Species:** Electrostatics not captured by 2D topology.
3. **Fused Rings:** Complex conjugation needs deeper architecture.
4. **Stereochemistry:** 3D geometry effects missing.
5. **Rare Groups:** Underrepresented in training.

8.2 Percentile Analysis

- Median: 0.049 eV
- 75th: 0.07 eV
- 90th: 0.15 eV
- 95th: 0.20 eV

Interpretation: 75% suitable for screening, top 5% need DFT refinement.

Systematic vs Stochastic: No systematic bias; stochastic errors manageable via uncertainty quantification.

8.3 Uncertainty Quantification

Future work:

- Monte Carlo dropout
- Deep ensembles (5-10 models)
- Conformal prediction intervals

9 STRENGTHS

9.1 Scale and Rigor

- ~400k molecules, sufficient statistical power
- Murcko split: gold standard evaluation
- Memory-safe: 16-32 GB RAM

9.2 Architecture

- GINE: edge-aware, efficient
- Dual pooling: extensive + intensive properties
- 6 layers: balances depth/oversmoothing

9.3 Training

- AMP + AdamW: stable convergence
- Reproducible: seed=42
- 0.008 eV gap: exceptional generalization

9.4 Applicability

- 1,000+ mol/s throughput
- Sub-0.1 eV MAE
- Interpretable error analysis

10 LIMITATIONS

10.1 2D Topology

Missing: conformational flexibility, stereochemistry, non-covalent interactions, solvent effects.

Impact: Cannot distinguish conformers with different gaps.

10.2 Limited Features

Missing: bond lengths, angles, dihedrals, partial bond orders.

Impact: Geometric constraints unencoded.

10.3 Single-Task

Training only on gap misses synergies with IP, EA, dipole, polarizability.

10.4 Test Distribution

Test MAE < Val MAE suggests easier test set.

Mitigation: Multiple splits with different seeds.

10.5 No Uncertainty

Point predictions only; cannot flag unreliable predictions.

Mitigation: Implement ensembles or conformal prediction.

10.6 Domain Limits

Trained on: 0–10 eV, 10–75 atoms, organic molecules.

Risk: May fail on organometallics, >100 atoms, exotic elements.

11 FUTURE WORK

11.1 3D Architectures

SchNet/DimeNet++/GemNet: Incorporate distances, angles, torsions.

Benefit: Improved stereochemistry handling.

11.2 Multi-Task Learning

Joint prediction: HOMO, LUMO, gap, IP, EA, dipole, polarizability.

Benefit: Shared representations improve generalization.

11.3 Active Learning

Select high-uncertainty samples for DFT labeling:

```
1 uncertainty = predictions.std(dim=0)
2 select_indices = uncertainty.topk(k=1000).indices
```

Benefit: Iterative improvement on challenging chemistries.

11.4 Transfer Learning

Pre-train on ChEMBL/ZINC with self-supervision, fine-tune on quantum data.

Benefit: Better generalization to underrepresented chemistries.

11.5 Explainability

Grad-CAM, subgraph attribution for interpretability.

Benefit: Chemist trust, structure-property insights.

12 CONCLUSION

This work demonstrates GINE achieves competitive HOMO-LUMO gap prediction on ~400k molecules with rigorous scaffold evaluation.

Key Contributions:

1. **Rigorous Protocol:** Murcko split + memory-safe preprocessing
2. **Performance:** MAE=0.069 eV, R^2 =0.9971, median=0.049 eV
3. **Scalability:** ~400k graphs on consumer hardware
4. **Analysis:** Comprehensive visualization and error profiling

Implications: Enables rapid molecular screening, accelerates materials/drug discovery.

Future: 3D coordinates, multi-task learning, uncertainty quantification will enhance reliability.

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